# Homework 1: Arc 1 Challenge

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# Problem Statement

The problem presents several pairs of input and output grids (usually 30x30 or smaller) demonstrating a transformation. The objective is to deduce the underlying logic and apply it to a new "test" input grid to generate the correct output.

We are addressing this challenge using a Diffusion-based approach. We provide a task file to the model, and it learns the underlying rule by denoising and subsequently predicting the correct output for a given input.

# Preprocessing

We have 900 task files.

- Each task file contains 3 train and 1 test example.
- We performed data augmentation using:
  - Rotations/reflections and transpose
  - Color shifting
- This resulted in 240 train examples and 80 test examples per task.
- Total of 50 * (240 + 80) *18 = 288,000 examples.

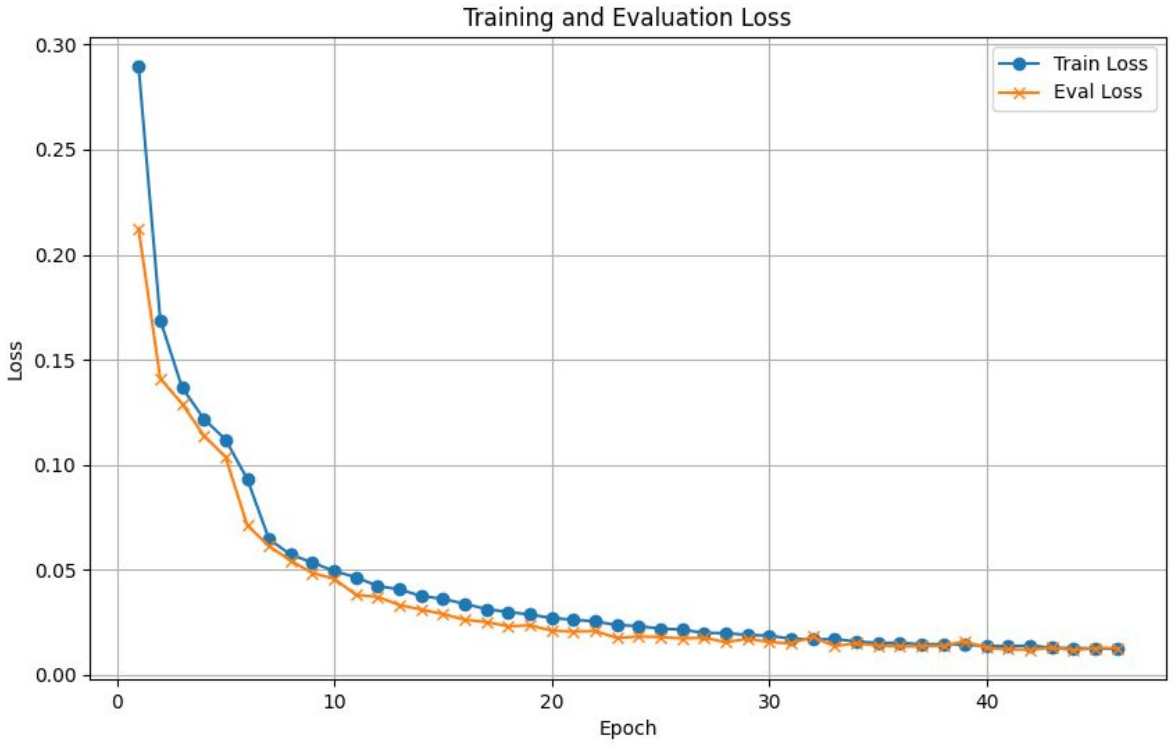
# Method

- The approach under consideration is a masked diffusion process.
- This uses data of discrete grid tokens (0-9).
- **Discrete Noise Scheduler:**
  - Adds random color values (0-9) as noise over t time steps to the output vector.
  - Noise addition stops at t_max, at which point the data is random noise.
- **Transformer-based Denoiser:**
  - Uses a transformer-based encoder for the diffusion process.
  - The transformer takes the input grid and the noisy output vector.
  - By using the input grid, the model learns a mechanism to map input constraints to denoise the output vector.

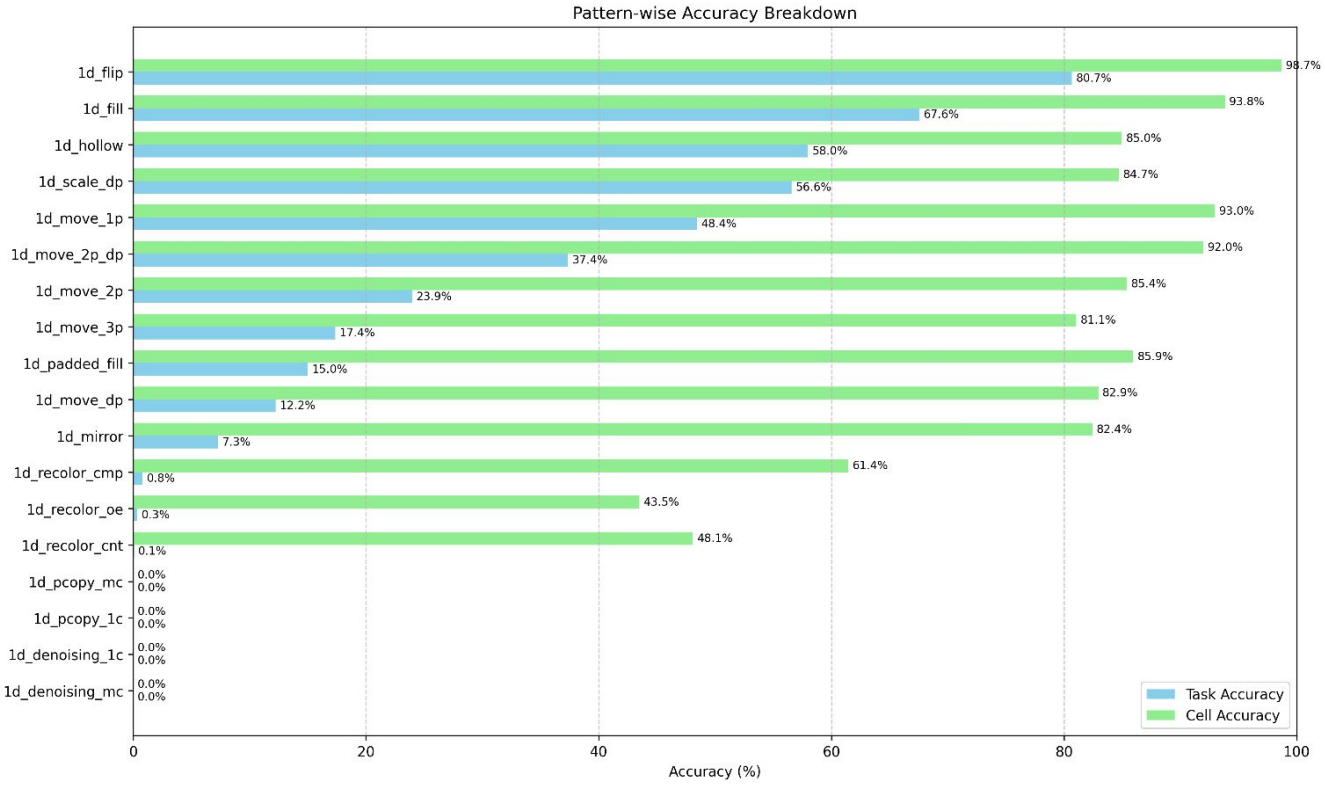
# Method

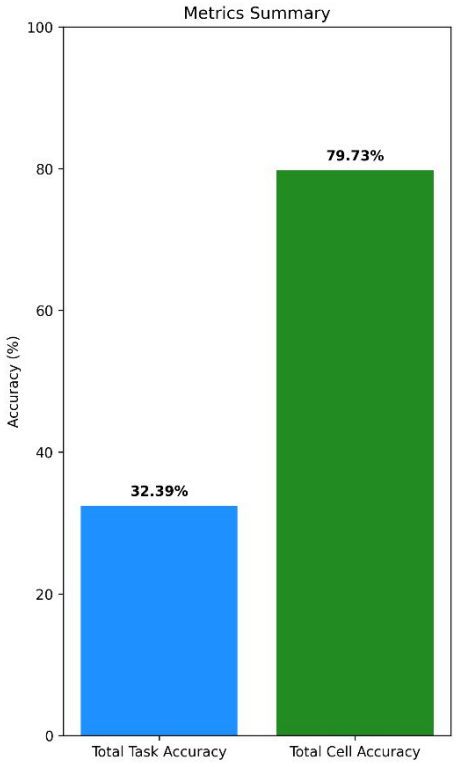
- The model predicts a rough initial grid in the first forward pass (no gradient).
- A second forward pass uses the initial guess and input grid to denoise the output vector, calculate error, and update the model via backpropagation.
- The training objective is to learn the original grid prediction by minimizing the cross-entropy loss to achieve the correctly predicted output grid.
- After training, the model predicts the output through iterative sampling using the input grid until convergence.

# Loss Curves


Training and Evaluation Loss
Train Loss
Eval Loss
Loss
Epoch
0.30
0.25
0.20
0.15
0.10
0.05
0.00
0
10
20
30
40

# Results


Pattern-wise Accuracy Breakdown
1d_flip
1d_fill
1d_hollow
1d_scale_dp
1d_move_1p
1d_move_2p_dp
1d_move_2p
1d_move_3p
1d_padded_fill
1d_move_dp
1d_mirror
1d_recolor_cmp
1d_recolor_oe
1d_recolor_cnt
1d_pcopy_mc
1d_pcopy_1c
1d_denoising_1c
1d_denoising_mc
98.7%
80.7%
93.8%
67.6%
85.0%
58.0%
84.7%
56.6%
93.0%
48.4%
92.0%
37.4%
85.4%
23.9%
81.1%
17.4%
85.9%
15.0%
82.9%
12.2%
82.4%
7.3%
61.4%
0.8%
43.5%
0.3%
48.1%
0.1%
0.0%
0.0%
0.0%
0.0%
0.0%
0.0%
0.0%
0.0%
Task Accuracy
Cell Accuracy
0
20
40
60
80
100
Accuracy (%)


Metrics Summary
100
80
60
40
20
0
Accuracy (%)
79.73%
32.39%
Total Task Accuracy
Total Cell Accuracy

# Homework 2: Stock Prediction and Portfolio Optimization

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# Problem statement

Develop a system to predict stock returns using historical data and efficiently allocate capital across tickers to maximize gains.

- **Data Source:** Nifty 50 daily OHLCV data from 2000 to 2026.
- **Approach:** A two-model system is employed:
  1. One model predicts individual stock returns based on past data.
  2. A second model determines portfolio allocation among the 50 stocks.

# Preprocesing

## Dataset Statistics

- Data includes 50 stocks, starting from the year 2000.
- The data is split into 80% training and 20% testing sets, applied on a monthly basis.

```
Total Train Samples: 171,001
Total Test Samples:  41,803
Total Unique Tickers: 44
Train Date Range: 2000-02-15 to 2026-03-16
Test Date Range:  2000-02-18 to 2026-03-11
```

| Ticker | Train Samples | Test Samples | Total Samples |
|---|---|---|---|
| ADANIENT | 4630 | 1133 | 5763 |
| ADANIPORTS | 3627 | 891 | 4518 |
| APOLLOHOSP | 4584 | 1120 | 5704 |
| ASIANPAINT | 4584 | 1120 | 5704 |
| AXISBANK | 4598 | 1124 | 5722 |
| BAJAJ-AUTO | 3533 | 866 | 4399 |
| BAJAJFINSV | 3532 | 866 | 4398 |
| BAJFINANCE | 4583 | 1120 | 5703 |
| BEL | 4584 | 1120 | 5704 |

| Ticker | Row Count | Start Date | End Date | First Close | Last Close |
|---|---|---|---|---|---|
| RELIANCE | 6487 | 2000-02-15 | 2026-03-16 | 31.60 | 1395.10 |
| TMPV | 6441 | 2000-02-15 | 2026-03-16 | 33.15 | 314.40 |
| ADANIENT | 5763 | 2003-01-01 | 2026-03-16 | 72.15 | 1978.00 |
| BHARTIARTL | 5722 | 2003-01-01 | 2026-03-16 | 11.10 | 1788.80 |
| HDFCBANK | 5722 | 2003-01-01 | 2026-03-16 | 10.80 | 840.60 |
| ONGC | 5722 | 2003-01-01 | 2026-03-16 | 33.25 | 260.45 |
| SUNPHARMA | 5722 | 2003-01-01 | 2026-03-16 | 13.36 | 1783.20 |
| TATACONSUM | 5722 | 2003-01-01 | 2026-03-16 | 17.43 | 1092.70 |
| SHRIRAMFIN | 5722 | 2003-01-01 | 2026-03-16 | 2.30 | 992.00 |
| SBIN | 5722 | 2003-01-01 | 2026-03-16 | 26.84 | 1066.70 |
| KOTAKBANK | 5722 | 2003-01-01 | 2026-03-16 | 8.48 | 369.85 |
| ITC | 5722 | 2003-01-01 | 2026-03-16 | 14.57 | 308.25 |
| M&M | 5722 | 2003-01-01 | 2026-03-16 | 14.14 | 3036.10 |
| LT | 5722 | 2003-01-01 | 2026-03-16 | 32.81 | 3469.40 |
| INFY | 5722 | 2003-01-01 | 2026-03-16 | 74.99 | 1249.80 |
| ICICIBANK | 5722 | 2003-01-01 | 2026-03-16 | 25.53 | 1272.90 |
| TATASTEEL | 5722 | 2003-01-01 | 2026-03-16 | 8.90 | 186.94 |
| AXISBANK | 5722 | 2003-01-01 | 2026-03-16 | 8.72 | 1214.70 |
| TITAN | 5722 | 2003-01-01 | 2026-03-16 | 3.87 | 4090.80 |
| TRENT | 5721 | 2003-01-01 | 2026-03-16 | 15.64 | 3596.10 |
| WIPRO | 5721 | 2003-01-01 | 2026-03-16 | 27.45 | 195.11 |
| HINDUNILVR | 5704 | 2003-01-27 | 2026-03-16 | 178.95 | 2175.70 |
| ASIANPAINT | 5704 | 2003-01-27 | 2026-03-16 | 22.10 | 2217.70 |
| HCLTECH | 5704 | 2003-01-27 | 2026-03-16 | 18.85 | 1328.60 |
| HINDALCO | 5704 | 2003-01-27 | 2026-03-16 | 51.86 | 921.15 |
| BEL | 5704 | 2003-01-27 | 2026-03-16 | 1.85 | 429.50 |
| APOLLOHOSP | 5704 | 2003-01-27 | 2026-03-16 | 52.45 | 7490.00 |
| CIPLA | 5704 | 2003-01-27 | 2026-03-16 | 66.38 | 1300.00 |
| GRASIM | 5704 | 2003-01-27 | 2026-03-16 | 55.30 | 2654.40 |
| EICHERMOT | 5704 | 2003-01-27 | 2026-03-16 | 8.15 | 6827.50 |
| BAJFINANCE | 5703 | 2003-01-27 | 2026-03-16 | 0.50 | 878.15 |
| MARUTI | 5593 | 2003-07-09 | 2026-03-16 | 164.30 | 12757.00 |
| ULTRACEMCO | 5331 | 2004-08-24 | 2026-03-16 | 260.20 | 11099.00 |
| TCS | 5330 | 2004-08-25 | 2026-03-16 | 121.60 | 2409.20 |
| NTPC | 5280 | 2004-11-05 | 2026-03-16 | 57.88 | 382.40 |
| JSWSTEEL | 5186 | 2005-03-23 | 2026-03-16 | 39.02 | 1145.60 |
| TECHM | 4845 | 2006-08-28 | 2026-03-16 | 69.28 | 1339.50 |
| POWERGRID | 4556 | 2007-10-05 | 2026-03-16 | 26.52 | 297.75 |
| ADANIPORTS | 4518 | 2007-11-27 | 2026-03-16 | 96.29 | 1373.90 |
| BAJAJ-AUTO | 4399 | 2008-05-26 | 2026-03-16 | 151.19 | 9073.00 |
| BAJAJFINSV | 4398 | 2008-05-26 | 2026-03-16 | 25.00 | 1773.90 |
| DRREDDY | 4008 | 2010-01-04 | 2026-03-16 | 114.22 | 1276.90 |
| NESTLEIND | 4004 | 2010-01-08 | 2026-03-16 | 62.35 | 1213.40 |
| COALINDIA | 3797 | 2010-11-04 | 2026-03-16 | 163.65 | 460.30 |
| INDIGO | 2561 | 2015-11-10 | 2026-03-16 | 438.62 | 4222.10 |
| SBILIFE | 2094 | 2017-10-03 | 2026-03-16 | 178.32 | 1909.20 |
| HDFCLIFE | 2063 | 2017-11-17 | 2026-03-16 | 114.87 | 626.00 |
| MAXHEALTH | 1383 | 2020-08-21 | 2026-03-16 | 111.75 | 970.60 |
| ETERNAL | 1154 | 2021-07-23 | 2026-03-16 | 126.00 | 222.04 |
| JIOFIN | 639 | 2023-08-21 | 2026-03-16 | 248.90 | 234.35 |

Total Unique Stocks: 50
Total Rows: 249863

# Preprocesing

- Making new features using OHLCV data to understand market context (bullish, bearish, or neutral) for a given sequence of N days.
- We are using two datasets: one for the stock return prediction model and one for the portfolio allocation model.
- All features are normalized before training for stability.

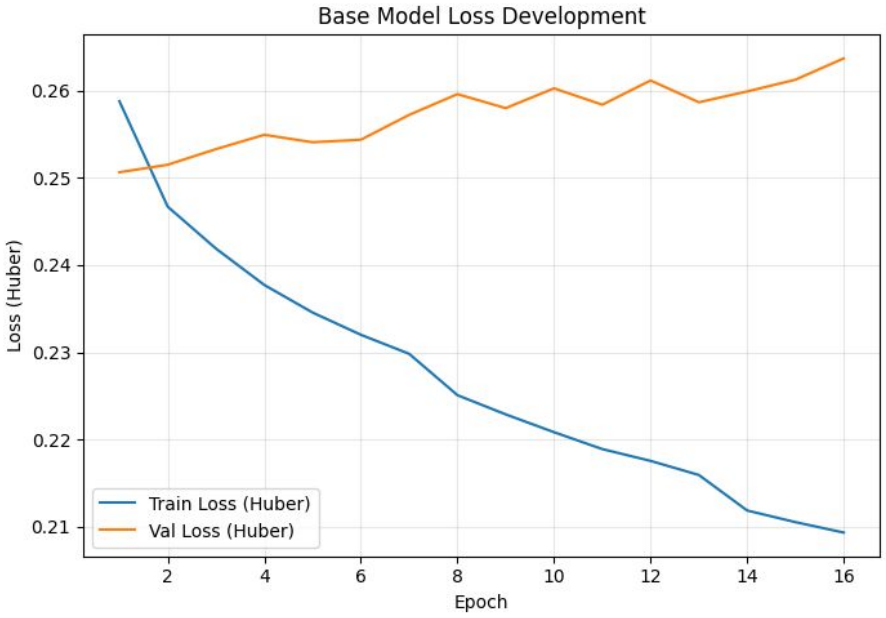
```
Using device: cuda
Reloaded uncontaminated internal statistics for
Loading portfolio data from ./data/data_50_stoc
Total structured trading days: 609
Train batches: 4 | Test batches: 1
```

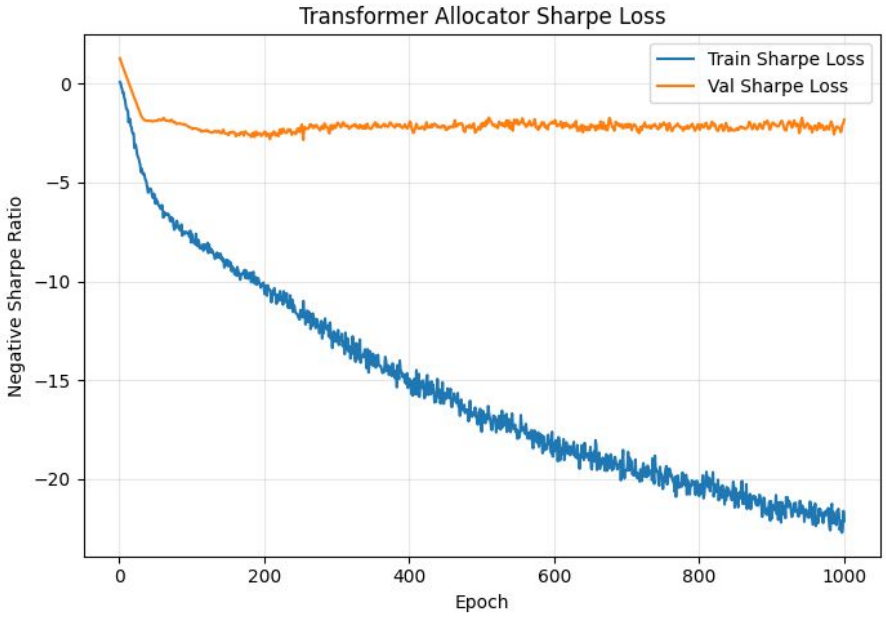
# Method

We are tackling the challenge of stock return prediction using a Convolutional Neural Network (CNN) that leverages historical data.

- The CNN was trained using the Huber loss function.
- For portfolio allocation, we employed a Transformer encoder model.
- During the training of the portfolio model, the weights of the stock return model were frozen, and its predictions were used to inform the portfolio allocations.
- These predicted returns guide the allocation for 50 different stocks.
- The training process for the portfolio allocation model is optimized by maximizing the Sharpe ratio.

# Loss Curves


Base Model Loss Development
Loss (Huber)
0.26
0.25
0.24
0.23
0.22
0.21
2
4
6
8
10
12
14
16
Epoch
Train Loss (Huber)
Val Loss (Huber)


Transformer Allocator Sharpe Loss
Negative Sharpe Ratio
0
−5
−10
−15
−20
0
200
400
600
800
1000
Epoch
Train Sharpe Loss
Val Sharpe Loss

# Results

```
Loading best model weights from model.pth...

Starting detailed per-stock evaluation for 44 stocks...
================================================================

[ADANIENT] - Performance for Date Range (Showing Last 10 evaluated days)
      Date  Ground Truth Return (%)  Predicted Return (%)  Absolute Error (%)
2025-12-19                  45.9429               16.6921             29.2508
2025-12-29                  77.9809               16.3967             61.5843
2026-01-09                  20.4173               14.0163              6.4010
2026-01-12                  57.5040               10.4261             47.0778
2026-01-16                -165.1208                9.5198            174.6406
2026-01-22                 -44.2171                7.9903             52.2074
2026-02-09                  -7.8430               -6.0386              1.8044
2026-02-13                -103.6355                2.1517            105.7871
2026-02-19                -176.8431                9.2060            186.0491
2026-02-26                 131.6628                3.3808            128.2820

[ADANIPORTS] - Performance for Date Range (Showing Last 10 evaluated days)
      Date  Ground Truth Return (%)  Predicted Return (%)  Absolute Error (%)
2025-12-19                  86.3244                5.4782             80.8463
2025-12-29                  74.6114               22.7539             51.8574
2026-01-09                 -49.2989               24.7222             74.0211
2026-01-12                  59.1946               24.1461             35.0485
2026-01-16                -174.4949               24.5085            199.0035
...
```

```
Evaluating directional accuracy on test data...

==============================================
  Test Directional Accuracy    : 61.64%
  Total Valid Evaluated Samples: 6100
==============================================
```

```
Running Transformer inference for random dates in the test set...

============= Tx Inference for Date: 2026-02-13 =============
================ Tx PnL SUMMARY for 2026-02-13 ================
 Transformer Model Simulated Return:     +0.59603
 Equal-Weight Baseline (Market Avg):     +0.05754
 Perfect Foresight (Best Single Stock):  +1.45465
==============================================================

--- TOP 10 ALLOCATED STOCKS (Highest Conviction) ---
    Ticker  Base_Predicted_Signal  Tx_Allocated_Weight  True_NextDay_Return  Actual_Return_Contribution
       TCS               0.055110             0.324159             0.525446                    0.170328
  HDFCBANK               0.027273             0.281549             0.164153                    0.046217
     TECHM               0.136092             0.182738             0.667967                    0.122063
      INFY               0.059735             0.155262             1.244148                    0.193169
   HCLTECH               0.112220             0.038292             1.232592                    0.047198
     WIPRO               0.045625             0.013746             1.060085                    0.014572
  HDFCLIFE               0.012923             0.004240             0.586672                    0.002488
HINDUNILVR               0.020859             0.000012            -0.111927                   -0.000001
  HINDALCO               0.067717             0.000001            -0.723840                   -0.000000
 COALINDIA               0.071960             0.000000            -0.188134                   -0.000000

--- BOTTOM 5 STOCKS (Least Allocated) ---
    Ticker  Base_Predicted_Signal  Tx_Allocated_Weight  True_NextDay_Return  Actual_Return_Contribution
 MAXHEALTH              -0.063838             0.000000            -0.249462                   -0.000000
     TITAN              -0.047373             0.000000             1.051092                    0.000000
SHRIRAMFIN              -0.080202             0.000000            -0.707058                   -0.000000
BAJAJ-AUTO              -0.051128             0.000000             1.138443                    0.000000
   DRREDDY              -0.094418             0.000000             0.430355                    0.000000
--------------------------------------------------------------
```